

TABLE OF CONTENTS

TABLE OF CONTENTS	1
1 - OVERVIEW	2
2 – TOOLS AND MANPOWER REQUIRED	3
3 – LOCKOUT AND TAGOUT	3
4 – DISASSEMBLY AND RF COIL REMOVAL	4
5 - REPLACING TWIN RF BODY COIL.....	7
5-1 Unpacking Twin RF Body Coil.....	7
6 – INSTALLING THE REPLACEMENT TWIN RF COIL	10
7 – RESTORATION	12
8 - PREPARING FAILED TWIN RF BODY COIL FOR SHIPMENT	14
REVISION HISTORY	15

1 - OVERVIEW

The Twin 1.5T RF body coil cartridge is a field-replacable unit (FRU) that mounts inside the gradient body coil (also a FRU). If the RF coil fails, it must be removed from within the gradient body coil and replaced. See Illustration 1-1.



REAR VIEW OF 1.5T TWIN RF COIL REMOVED FROM INSIDE GRADIENT COIL
ILLUSTRATION 1-1

Since the replacement procedures for the RF and gradient body coils are closely related, references are made below to the epoxy-filled gradient coil replacement procedure where the instructions for removal of alike parts (bridge removal, for example) can be found.

While some of the following procedure can be accomplished by one person (connecting/disconnecting various cables, etc.), replacing a failed RF body coil is a job for three people.

NOTE

Areas of text underlined and colored in [blue](#) signify text that is hyper-linked to documents on the 8.X Service Methods CDROM. As long as the CDROM drive is designated on the laptop as drive E, clicking on this hyper-linked text should allow the reader to immediately view the document.

2 – TOOLS AND MANPOWER REQUIRED

- Non-magnetic Tool Kit (GE part # 46-320273G4)
- Gradient chiller coolant (GE part # 2138791)
- 3 people required

3 – LOCKOUT AND TAGOUT



POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE MR SYSTEM OR RF CABINET BEFORE ATTEMPTING TO SERVICE THE TWIN RF BODY COIL. PERFORM THE LOCKOUT AND TAGOUT PROCESS TO THE APPROPRIATE CIRCUIT BREAKERS ON THE FRONT OF THE PDU TO SAFELY REMOVE POWER FROM THE RF BODY COIL CIRCUITS.

1. Refer to [SECTION 8 – MR SYSTEM LOCKOUT PROCEDURES](#) in the Safety document (SYSMSA2.DOC) located on the 8.X Service Methods CDRom.
2. Locate the subsection referring to the PDU used in the MR system.
3. Perform the Lockout and Tagout procedures described in that section.

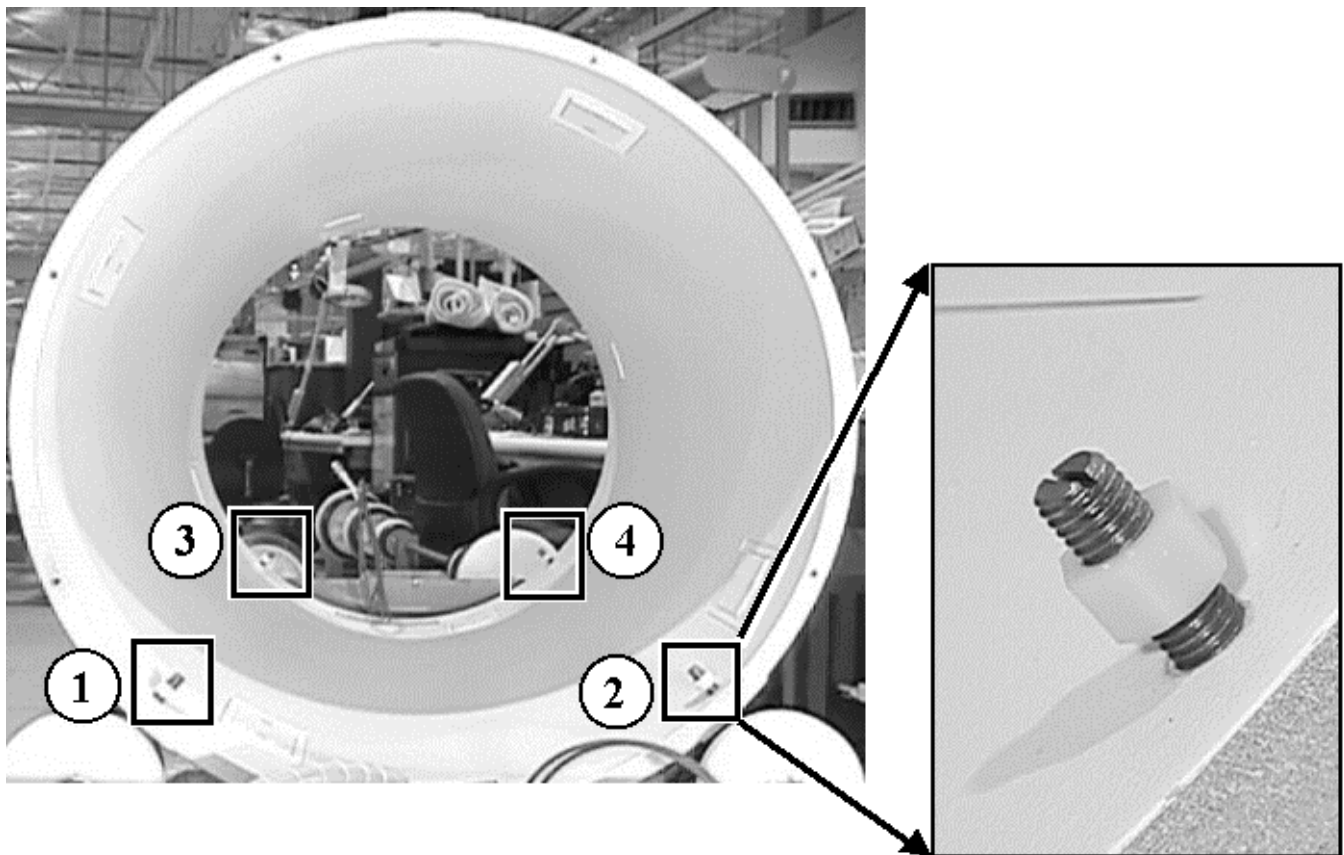
4 – DISASSEMBLY AND RF COIL REMOVAL

1. Perform **SECTION 3 – LOCKOUT AND TAGOUT** (Refer to *MR Signa 8.X Service Methods CDROM*, Safety (SYSMSA2.DOC), [Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS](#).).
2. Remove the front bridge cover
 - a. Remove the two (2) M6 Phillips screws securing the cover to the front bridge
 - b. Remove the front bridge cover and set aside in a safe location out of the way of traffic.
3. Detach the two parts of the split bridge by removing the four (4) M10 nuts holding the two sections together. Then, slide the rear part of the bridge by moving the rear pedestal until the two halves of the bridge are completely disengaged.
4. Detach the front bridge support from the front end bell.
 - a. Loosen the lock nuts at the bottom of the support legs.
 - b. Remove the two (2) M10 bolts using an adjustable wrench or a 17mm socket wrench.

NOTE

It is not necessary to detach the bridge support from the bridge.

5. Slide the long section of the bridge out of the magnet bore.
6. Turn off the gradient cooling lines (inlet and outlet) at the valve located in front of the magnet.
7. Set the SCC to “Service Mode” by flipping the toggle switch from the top position to the bottom position. This will prevent the Magnet Monitor from dialing out.
8. Turn System Cooling Cabinet (SCC) off and perform lock-out/tag-out.
9. Remove both coolant lines from the coil and let drain into a bucket or container.
10. Disconnect the coaxial cables from the RF coil.
 - a. Remove the two (2) coaxial bias lines labeled J3/J5 and J4/J6 at the front and rear of the RF coil.
 - b. Remove the two (2) rigid, blue coaxial RF drive cables from the bulkhead at the rear of the RF coil.
11. Remove the front and rear cover straps located at the interface between the front and rear end bells and the RF coil.
12. Use a non-magnetic 5/16” slot screwdriver to extend the two (2) feet at the 4 and 8 o’clock positions at the bottom of each end of the RF coil (total of 4 feet) until each foot contacts the gradient coil underneath. See Illustration 4-1.



LOCATION OF RF COIL FEET
ILLUSTRATION 4-1

13. Remove the bore light caps at the 10 o'clock and 2 o'clock positions on the front end-bell.
 - a. Remove the M4 screws securing each cap to the front end-bell.
 - b. Remove the bore light caps and relocate to a safe location out of the way of traffic.
14. Carefully detach (peel) the bore lights from the RF tube surface.
15. Remove the bore temperature sensor cable from the groove in the front end-bell and detach it from the SRI.
16. Remove the four (4) M10 screws using a 8mm Allen wrench at predefined positions.
17. Replace the screws removed in step 14 by M10 studs.
18. Open access panels on the front end-bell 8 M6 screws (flat head screw driver) typical at two locations.

CAUTION

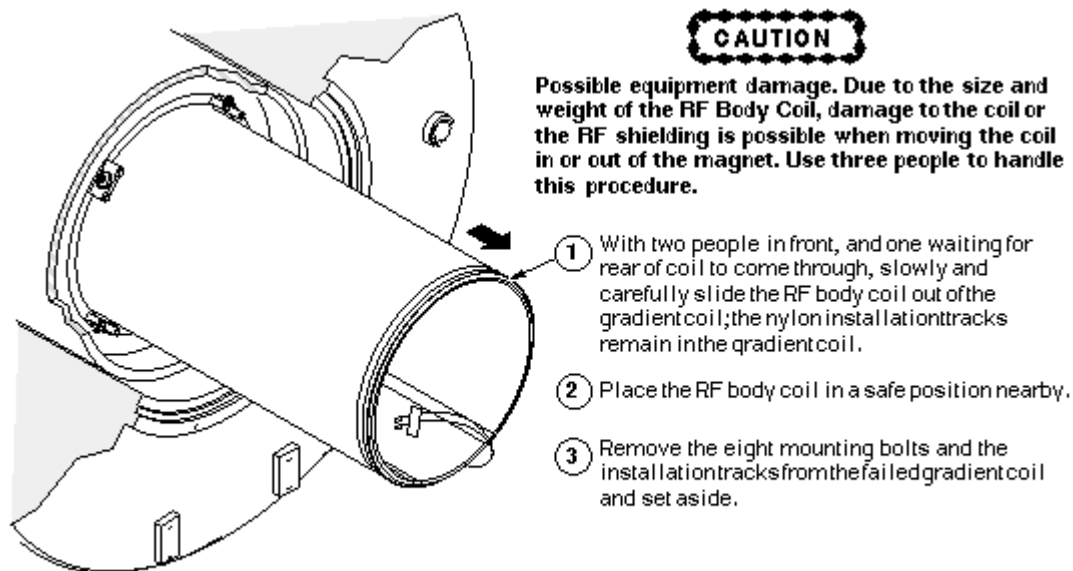
Coolant spillage may occur in the next step. Have absorbent towels available in case this happens.

19. Detach the 4 gradient coolant lines from the inside of the end-bell (Push connectors) and, to minimize spillage, fold the lines back onto themselves and temporarily secure into this position with rope-cord or adhesive tape.
20. Remove the 8 nylon nuts that secure the front end bell to the RF Coil.

21. Remove the 8 nylon nuts that secure the rear end bell to the RF Coil.
22. Remove the remaining twenty-four (24) M10 bolts on the front end bell, in a star pattern, using the 8mm Allen wrench.
23. Completely slide out the end-bell.
24. Refer to Illustration 4-2. Slide the RF coil out of the gradient coil.

NOTE

Ignore step 3 in Illustration 4-2 as there are no mounting bolts on the Twin RF coil.



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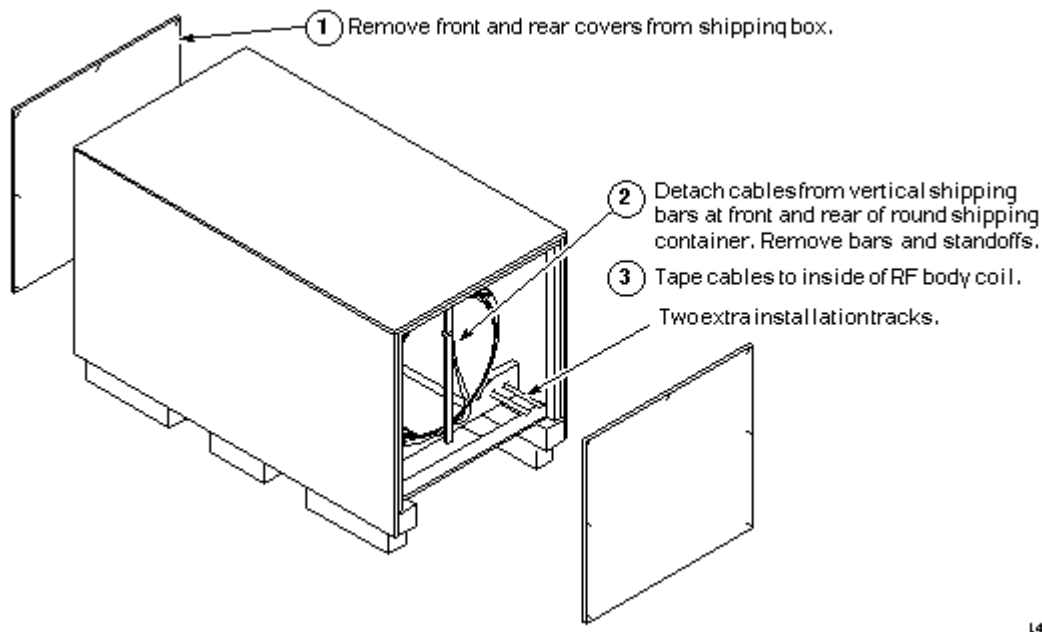
SLIDING OUT RF BODY COIL ON TRACKS
ILLUSTRATION 4-2

25. Carefully inspect gaskets on the front and rear end-bells. Confirm that the gasket surface that contacts the RF Coil is in good condition (not nicked, cut, severely deformed or otherwise damaged) and will not leak when subjected to a vacuum. Replace the gasket(s) if damage is found.

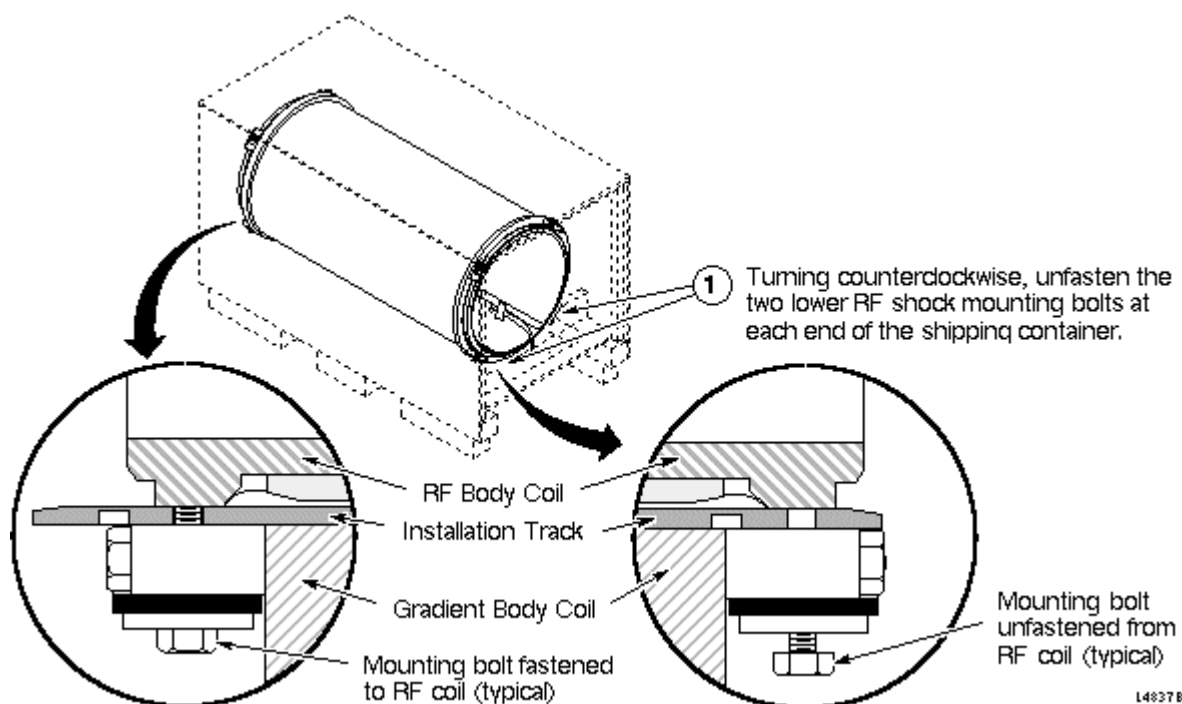
5 - REPLACING TWIN RF BODY COIL

5-1 Unpacking Twin RF Body Coil

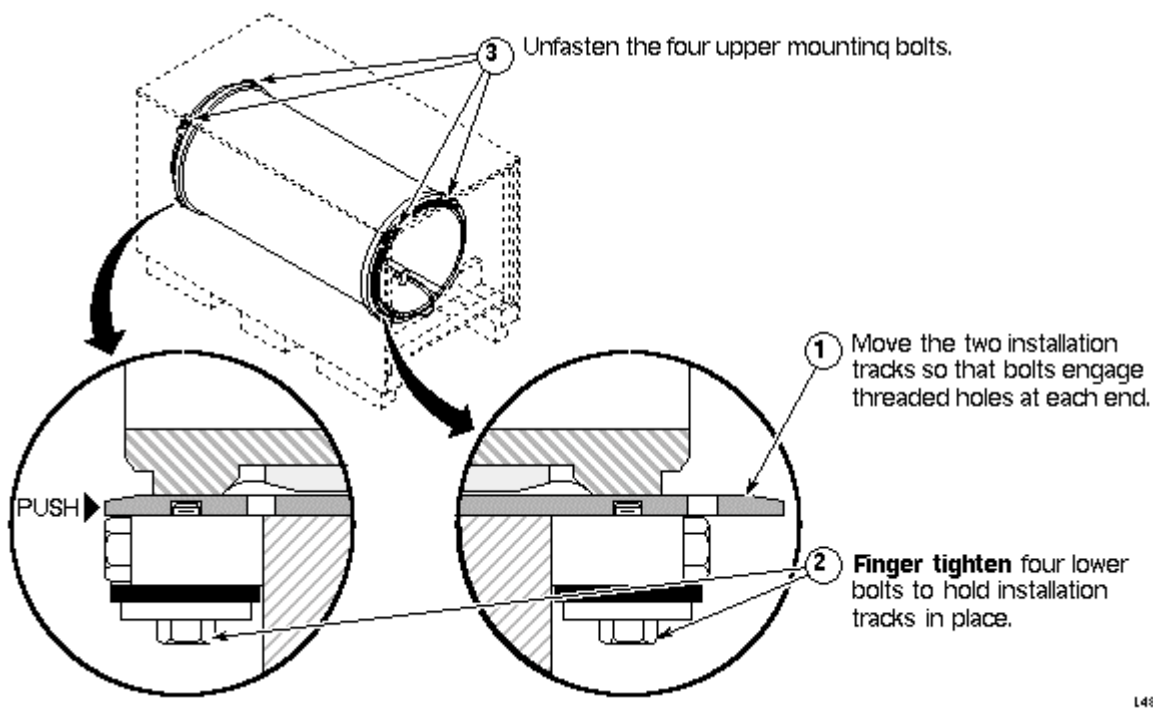
A new RF body coil is packed in a square shipping container that is strapped inside a combination pallet and outer shipping box. It rests on a pair of installation tracks. The shipping container never leaves the box; the RF body coil is removed from this combination. Included are two extra installation tracks that are used in the gradient body coil to help you make the RF coil change. To perform all necessary steps for this part of the procedure, work from Illustration 5-1 through Illustration 5-4.



UNPACKING THE NEW RF BODY COIL
ILLUSTRATION 5-1



SEPERATING RF BODY COIL FROM SHIPPING CONTAINER
ILLUSTRATION 5-2



SHIFTING INSTALLATION TRACKS IN SHIPPING CONTAINER
ILLUSTRATION 5-3

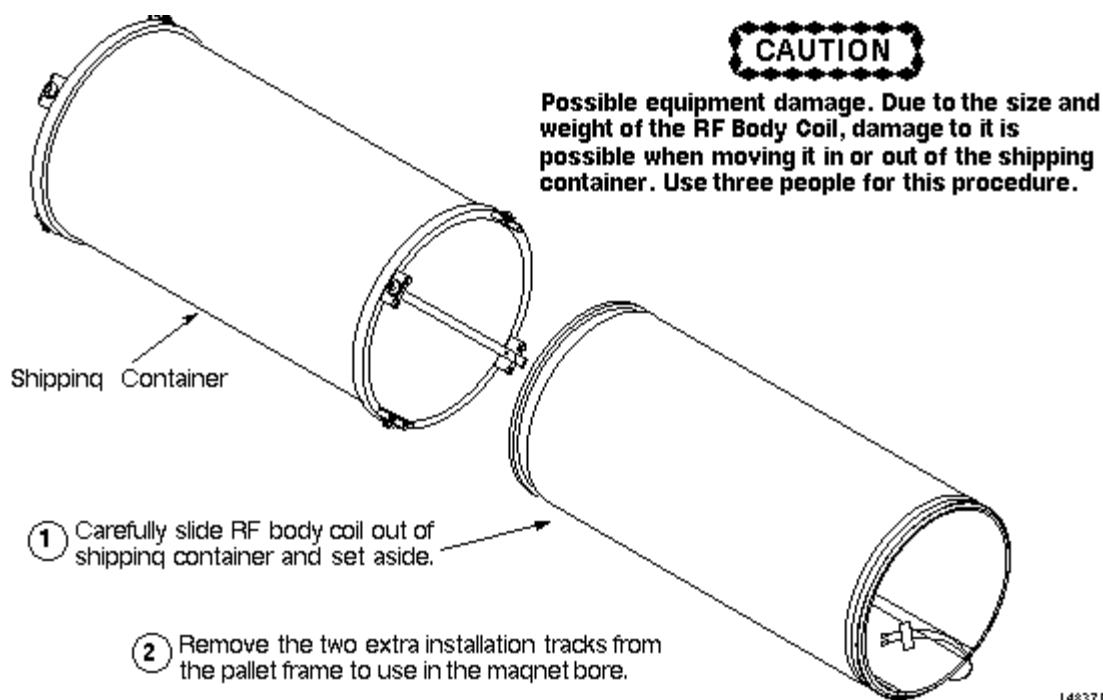
**REMOVING RF BODY COIL FROM SHIPPING CONTAINER**

ILLUSTRATION 5-4

NOTE

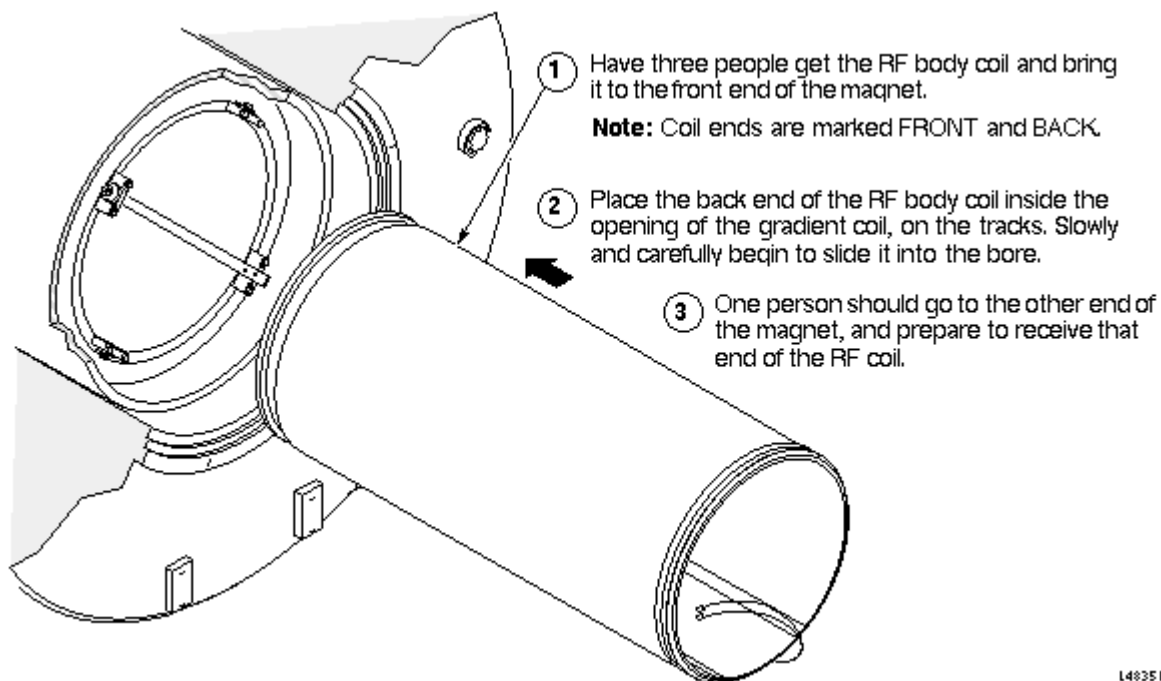
Installation tracks are not used to install or remove a Twin RF coil.

6 – INSTALLING THE REPLACEMENT TWIN RF COIL

1. Use a non-magnetic 5/16" slot screwdriver to extend the two (2) feet at the 4 and 8 o'clock positions at the bottom of each end of the replacement RF coil (total of 4 feet) about 0.3 inch (8 mm) from the outer diameter of the coil.
2. Confirm that the legs at the 12 o'clock position on each end of the RF coil are fully retracted and flush with the coil surface. These legs can only be adjusted with an Allen wrench.
3. Refer to Illustration 6-1 and install the replacement RF coil.

CAUTION

Use care to ensure that the legs on each end of the RF coil do not catch on the edge of the fiberglass straps in the center of the gradient coil or tear the reflective TRS (Thermal Radiation Shield) film exposed on the inside, front of the gradient coil.



SLIDING NEW RF COIL INTO PLACE
ILLUSTRATION 6-1

- a. Insert the coil into the bore until the RF coil studs pass through the slots in the rear end-bell surface and the RF coil surface contacts the rear end-bell.
 - i. Adjust the height of the rear legs to make the RF coil surface as level as possible with the rear end-bell surface.
 - ii. Roughly adjust the height of the front legs.
 - iii. Slightly rotate the RF coil as needed to until the rear RF coil studs pass through the slots on the rear end-bell. Rotational centering is not critical at this time.

4. Replace the front end-bell.
 - a. Attempt to place the front end-bell on the front of the RF coil so that studs match with the holes on the end-bell.
 - b. Adjust the height of the front legs, if necessary, in order to make the RF coil surface as level as possible with the front end-bell surface.
 - c. Seat the studs into the front end-bell and then rotate the end-bell and RF coil until the twenty-four (24) holes on the front end-bell match up with those on the magnet stainless-steel ring.
 - d. Install 8 nylon nuts to secure the front end bell to the RF Coil.
 - e. Install 8 nylon nuts to secure the rear end bell to the RF Coil.
 - f. Use a 8mm Allen wrench to install the twenty-four (24) M10 bolts in a star pattern on the end-bell and secure the end-bell to the enclosure.
5. From the front end-bell access panels reconnect the four (4) gradient coolant lines.
6. Replace the front end-bell access panels and secure using the eight (8) M6 screws.
7. Reinstall the bore temperature sensor cable into the groove in the front end-bell.
8. Coil the excess cable, leaving about 2 feet of the cable end free, and tie-wrap the cable to the studs securing the SRI to the magnet face.
9. Connect the end of the cable to J10 on the SRI.
10. Reinstall the bore lights onto the inside surface of the RF coil.
11. Reinstall the bore light caps using the two (2) M4 screws at the 10 o'clock and 2 o'clock positions on the front end-bell.
12. Fully retract the two (2) threaded front and rear RF coil legs (4 total) and then tighten the plastic nut until snug so that the legs do not vibrate. **DO NOT OVERTIGHTEN PLASTIC NUT.**
13. Reinstall the front and rear cover straps at the front and rear RF coil to end-bell interfaces.
14. Reconnect the coaxial cable bias and RF drive lines.
 - a. Reconnect the front and rear pair of coaxial cable bias lines labeled J3/J5 and J4/J6.
 - b. Reconnect the pair of rigid, blue coaxial cable I and Q RF drive lines at the rear of the RF coil to the connectors on the bulkhead interface.
15. Place the gradient coolant line valves behind the magnet in the open position.
16. Remove all lock-out/tag-out devices.

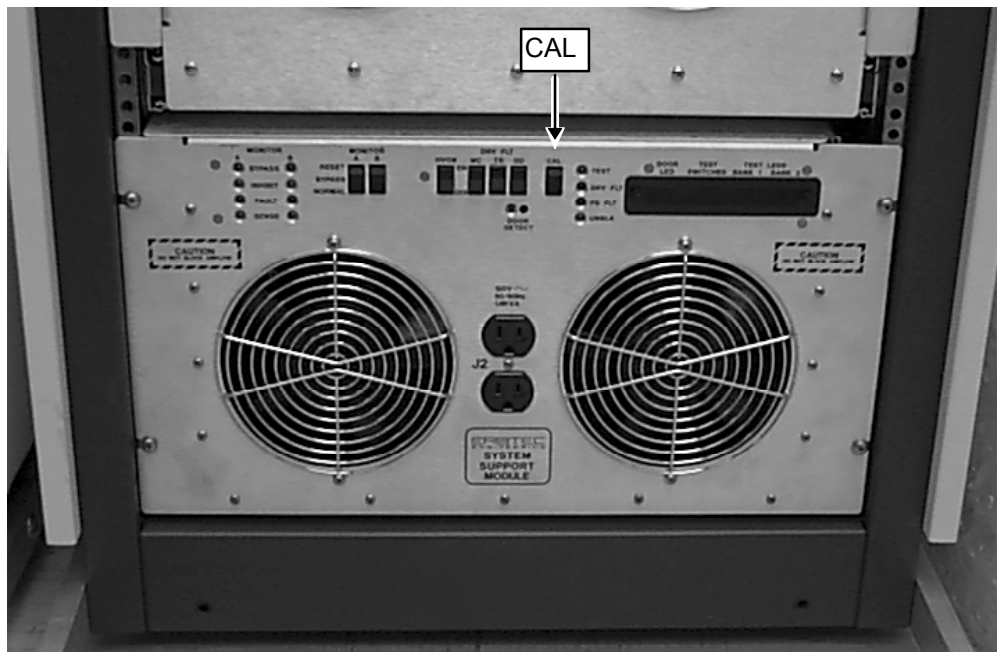
7 – RESTORATION

1. Refer to [SECTION 8 – MR SYSTEM LOCKOUT PROCEDURES](#) in the Safety document (SYSMSA2.DOC) located on the 8.X Service Methods CDROM.
2. Locate the subsection referring to the PDU used in the MR system.
3. Perform the Restoring Power section provided for the PDU used in the MR system.
4. Verify the gradient chiller is running, check the coolant level, and replenish coolant as necessary.
5. View the fittings through the transparent window on the front end-bell and confirm that none of the fittings are leaking coolant.
6. Check the gradient coolant valves behind the magnet and confirm that these are also not leaking.
7. Reinstall the bridge into the bore.
 - a. Slide the long section of the bridge back into the bore.
 - b. Secure the front bridge support to the end-bell using the two (2) M10 bolts.
 - c. Tight the locknuts on the M10 bolts.
8. If the system has a split bridge, move the rear pedestal until the two parts of the bridge are engaged. Secure with the four (4) M10 nuts and an adjustable wrench.
9. Replace front Aesthetic ring (no tools needed).
10. Replace front bridge cover and secure using two (2) M6 Phillips screws.
11. Confirm that LEDs are illuminated on the front of the RFI and SSM and confirm that the fans in the front of the RF amplifiers are rotating.

WARNING!

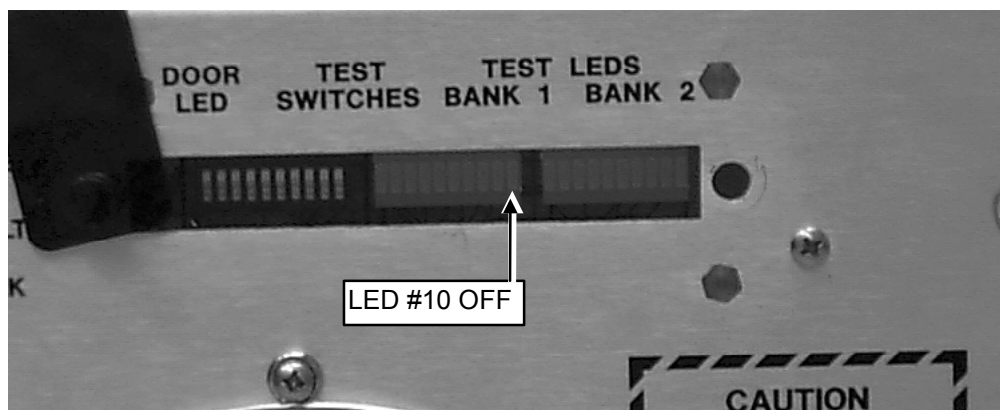
THE SSM WILL NOT FAULT IF IT IS CALIBRATED WITH A DISCONNECTION OR POOR CONNECTION TO ONE OF THE 4 MAGNET ROOM DYNAMIC DISABLE BIAS LINES (J72, J73, J75, AND J76 ON THE MAGNET ROOM SIDE OF THE PENETRATION PANEL). IMAGE SNR DEGRADATION WILL BE NOTICED BUT THE SSM MAY NOT FAULT. MAKE CERTAIN THE FOUR BIAS LINE CONNECTIONS FROM THE PEN PANEL TO THE RF COIL ARE CONNECTED BEFORE PERFORMING THIS CALIBRATION.

12. Press and release the CAL switch on the front of the SSM (see Illustration 7-1). This samples the open circuit fault detect signals and adjusts the threshold accordingly for the Dynamic Disable Circuits: J42 (Body 1), J43 (Body 2), and J44 (Direct Drive). When the CAL switch is pressed the LEDs labeled TEST, DRV FLT, and PS FLT will illuminate (this does not indicate a fault).



SSM CAL SWITCH LOCATION
ILLUSTRATION 7-1

13. View the front panel LEDs. Ensure the Bank 1, LED #10 is OFF (see Illustration 7-2). This LED indicates the adjustment process was successful.

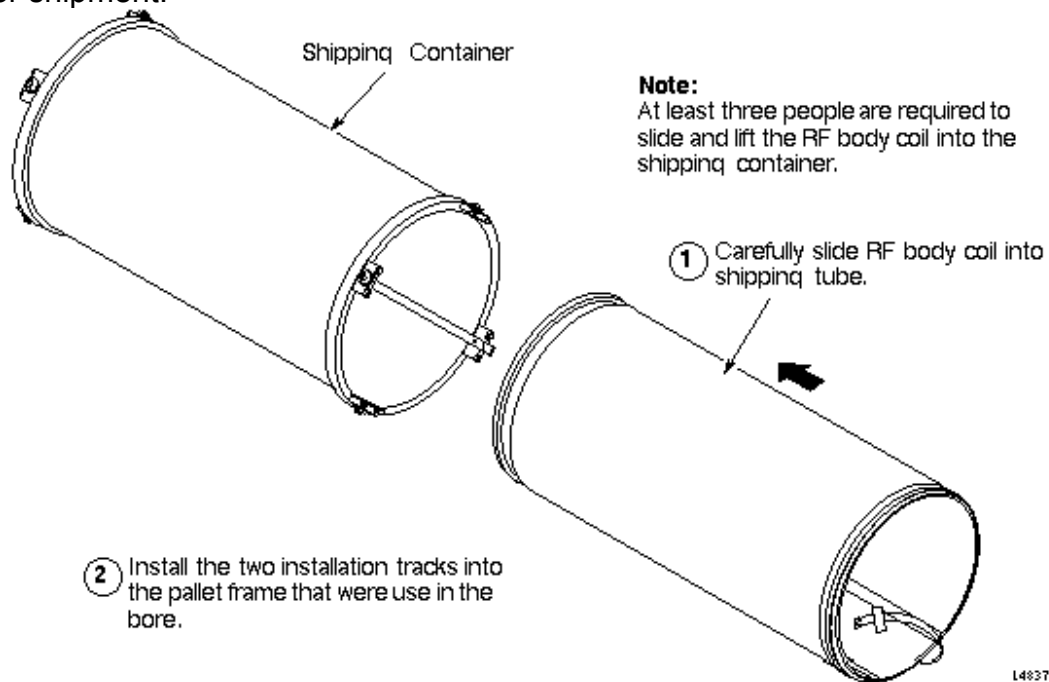


SSM FRONT PANEL LEDs
ILLUSTRATION 7-2

14. Verify that the system is functional by completing one head and body scan.
15. Dispose of excess gradient coolant drained earlier from the gradient coil as per local ordinances and regulations.
16. Flip the toggle switch on the SCC cabinet back to the upper position so the Magnet Monitor will be able to dial out.

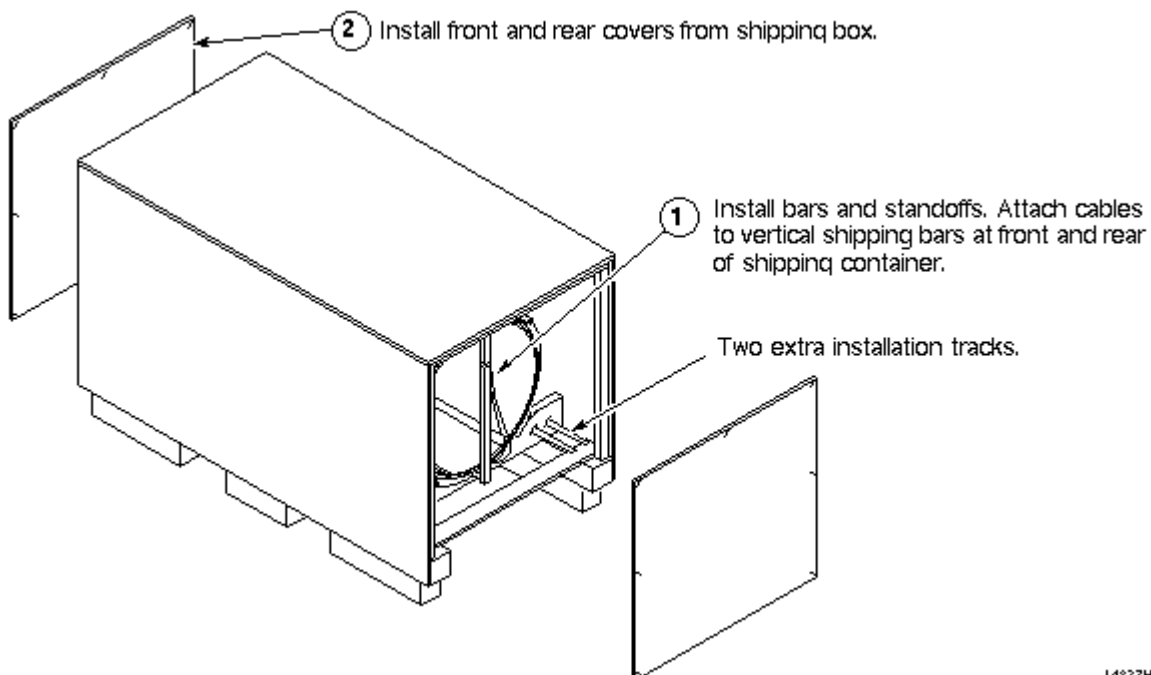
8 - PREPARING FAILED TWIN RF BODY COIL FOR SHIPMENT

Illustrations 8-1 through 8-2 provide instructions explaining the packing of the failed RF Body Coil for shipment.



FAILED RF COIL INTO SHIPPING CONTAINER
ILLUSTRATION 8-1

Stabilize the RF coil inside the container by extending all three (3) extension legs on the front and rear of the RF coil.



CLOSING BRM RF SHIPPING CONTAINER AND BOX
ILLUSTRATION 8-2

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	August 14, 2001	D. Thome'	Initial
1	Oct. 16, 2001	K. Schraufnagel	Updated coil removal steps.